

BAUDETTE, MN, USA

WMO: 727476

Lat: 48.717N

Lon: 94.600W

Elev: 1082

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94961

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.4	-19.5	-28.8	1.1	-22.2	-25.1	1.4	-17.9	24.5	8.8	22.2	7.1	3.8	310	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	85.2	70.5	82.3	68.4	79.9	66.6	73.1	82.0	71.0	79.7	68.8	76.6	11.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	115.4	78.9	67.9	106.6	76.2	65.8	98.9	74.0	37.4	82.1	35.4	79.8	33.6	76.7	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.8	19.1	17.2	-31.2	90.0	7.3	2.2	-36.4	91.6	-40.7	92.9	-44.8	94.1	-50.1	95.7		
(5)			-28.6	76.5	4.8	2.1	-32.1	78.0	-34.9	79.2	-37.6	80.3	-41.1	81.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.8	6.0	10.4	23.7	39.3	51.9	62.3	67.3	64.9	56.4	42.9	27.0	12.0		
	DBStd	23.87	14.82	13.94	13.77	10.18	8.36	6.64	5.39	5.74	8.09	8.64	11.50	13.51		
	HDD50	5862	1364	1109	818	339	77	3	0	0	28	255	692	1177		
	HDD65	9804	1829	1529	1280	770	412	126	39	73	277	686	1141	1642		
	CDD50	1784	0	0	3	20	136	372	536	462	219	35	1	0		
	CDD65	249	0	0	0	0	6	44	110	70	18	1	0	0		
	CDH74	2466	0	0	1	9	108	468	1010	667	189	14	0	0		
	CDH80	543	0	0	0	1	19	103	238	144	37	1	0	0		
	WSAvg	7.7	7.5	7.5	7.8	8.7	8.6	7.3	6.6	6.7	7.4	8.2	8.1	7.5		
	PrecAvg	22.5	0.6	0.4	0.7	1.3	2.7	4.0	3.4	3.3	2.6	2.0	0.9	0.6		
(15) Precipitation	PrecMax	30.0	2.6	1.3	3.5	3.9	6.1	14.5	6.5	7.8	9.1	7.6	3.7	1.7		
	PrecMin	14.5	0.0	0.0	0.0	0.0	0.5	0.9	1.1	0.7	0.4	0.2	0.0	0.0		
	PrecStd	4.0	0.5	0.3	0.6	0.9	1.5	2.3	1.2	1.8	1.7	1.5	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	43.0	63.8	74.0	82.9	88.1	89.4	88.1	84.3	75.0	59.4	40.3		
		MCWB	35.6	37.4	53.4	55.4	63.6	72.4	73.2	72.8	69.1	60.0	50.4	35.4		
	2%	DB	34.4	38.5	52.7	67.2	76.9	82.5	85.4	83.8	79.6	67.8	51.8	35.9		
		MCWB	32.1	34.5	44.4	50.3	59.3	67.7	71.0	69.6	65.9	56.2	44.8	33.1		
	5%	DB	31.0	35.0	45.9	61.9	72.5	79.4	82.4	80.9	74.9	62.1	46.2	33.4		
		MCWB	29.0	32.5	39.4	47.9	57.2	65.2	69.3	67.9	63.4	53.0	41.1	31.5		
	10%	DB	27.0	31.3	41.1	56.9	68.1	75.9	79.8	77.8	71.0	56.5	41.6	30.1		
		MCWB	25.3	29.1	36.3	45.1	55.2	63.6	67.6	65.7	61.7	49.3	38.1	28.4		
	0.4%	WB	35.9	38.6	54.0	56.3	67.5	74.5	76.0	75.2	72.0	62.3	51.4	36.2		
		MCDB	39.6	41.6	58.3	69.3	76.9	83.2	85.3	84.5	81.8	69.3	56.6	38.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.6	35.0	44.9	52.0	63.0	70.8	73.5	72.2	68.3	58.1	45.7	33.5		
		MCDB	34.6	37.9	51.9	63.6	71.0	80.5	81.8	80.9	75.4	67.4	50.4	35.3		
	5%	WB	29.1	32.5	39.9	49.2	59.7	67.9	71.4	69.6	65.5	53.5	41.7	31.4		
		MCDB	31.0	35.0	45.1	59.9	68.2	75.6	80.0	77.5	72.6	59.9	45.9	33.2		
	10%	WB	25.3	29.1	36.5	46.2	56.8	65.4	69.2	67.7	62.6	50.0	38.0	28.3		
		MCDB	27.0	31.3	40.9	56.4	66.1	72.9	77.0	75.4	69.3	55.8	41.5	30.0		
	5% DB	MDBR	18.3	20.0	20.1	20.6	21.1	20.0	20.9	21.2	20.0	16.5	14.0	15.1		
		MCDBR	17.5	19.3	22.6	29.9	28.6	25.6	24.2	24.8	26.1	26.1	20.9	15.8		
		MCWBR	15.3	15.5	15.1	17.2	14.5	13.0	12.5	12.7	14.3	15.6	14.3	13.6		
		MCWBR	17.0	16.9	21.6	26.7	23.3	21.5	21.6	21.6	21.5	23.9	19.1	15.1		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.268	0.305	0.349	0.358	0.375	0.380	0.375	0.351	0.305	0.278	0.253		
		taud	2.352	2.309	2.372	2.370	2.409	2.386	2.394	2.427	2.475	2.575	2.499	2.386		
	Ebn at Noon		264	284	290	285	286	281	278	274	271	270	253	248		
		Edh at Noon	21	28	32	35	36	36	36	33	29	22	19	18		
(44) All-Sky Solar Radiation	RadAvg		445	828	1207	1520	1705	1794	1925	1605	1189	700	397	321		
	RadStd		49	67	119	134	106	141	90	117	94	102	46	34		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-2.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page